

Two pathways, many clocks: a commentary on Itao (2026)

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Abstract

Itao (2026) reports two universal demographic-transition pathways: a conserved λe_0 before and a conserved $\lambda \exp(e_0/17)$ after a mid-twentieth-century change point. Using the same public data, we ask whether this is a placeless law or partly an artefact of aligning asynchronous national transitions on a common calendar, and whether it holds at sub-national scales. Country-specific onsets span 1828–2013 (IQR 58 years). Event-time alignment reduces between-country λ variance by a factor of 3.4, but tightens only the first conserved quantity. Within countries, regional λ at matched e_0 differs as much as the cross-national variation of the second pathway. The two pathways are a timing-locked regularity plus heterogeneous convergence rather than a single universal constant.

Keywords: demographic transition; fertility; life expectancy; universality; sub-national heterogeneity; event-time alignment

A cross-disciplinary claim

The search for law-like regularities in human populations has a long history (Notestein, 1945; Kirk, 1996). Itao (2026) contributes a striking recent example: plotting λ against life expectancy at birth e_0 for 186 countries, the paper reports that national trajectories are organised by two conserved quantities, λe_0 before and $\lambda \exp(e_0/17)$ after a change point near the mid-twentieth century. Importing statistical-physics search for conserved quantities into demography is a valuable cross-disciplinary move, and the household model linking the shift to returns to schooling (Galor, 2011) gives a plausible mechanism. We take the two-regime description as a starting point and ask how far the regularity survives when timing and spatial scale are treated more carefully.

Asynchronous transitions, not a global turning point

If the reported change point were a genuine world-historical moment, most countries should transition near it. They do not. Using Coale's (1973) onset definition—the first irreversible decline in a smoothed λ series—country-specific onsets range from 1828 to 2013, an inter-quartile range of 58 years; only 6% begin within a decade of 1950 (Figure 1A). A pooled calendar split is sharply estimated at 1966 (95% CI 1946–1969), consistent with Itao (2026), but this narrowness is an aggregation artefact: it summarises asynchronous national transitions rather than a synchronised global event (Bongaarts & Watkins, 1996; Reher, 2004).

Event alignment sharpens the first pathway only

If transitions are asynchronous, a placeless law should look clearer when each country is read on its own clock. Re-aligning trajectories on each country's onset reduces the between-country standard deviation of λ from 10.9 to 5.9 births per 1,000 in the ± 25 -year window—a 3.4-fold reduction in variance (Figure 1B). Under alignment, the first conserved quantity, λe_0 , tightens

from $CV = 0.27$ to 0.17 , a timing-locked regularity that calendar-time pooling partly obscures. The second quantity, $\lambda \exp(e_0/17)$, does not improve ($CV = 0.30$ versus 0.31 ; Figure 1C). The two pathways are asymmetric: the pre-transition process has a conservation-like structure once timing is modelled, whereas the post-transition regime remains a loose, heterogeneous convergence.

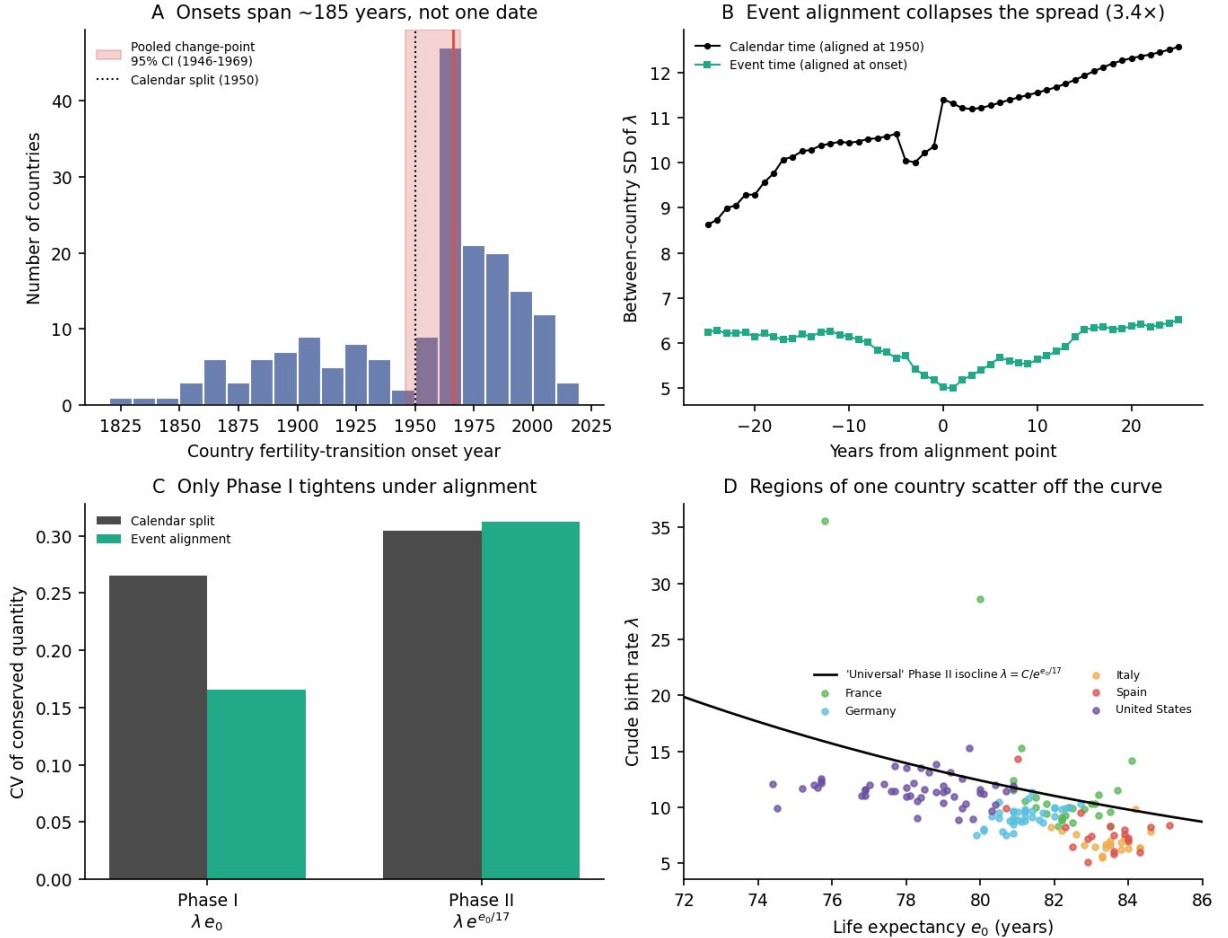


Figure 1. (A) Fertility-transition onsets span 185 years; the shaded band is the 95% CI for the pooled calendar change point. (B) Event-time alignment (each country at its own onset) reduces the between-country SD of λ by a factor of 3.4. (C) Alignment lowers the CV of the first conserved quantity (λe_0) but not the second ($\lambda \exp(e_0/17)$). (D) Within five decentralised states, regions at nearly identical e_0 scatter off the single 'universal' Phase II isocline, so the country-level curve does not carry to sub-national scales.

No single curve describes within-country variation

A genuinely placeless law should also hold below the national level. Region-level data for 156 regions of five decentralised states (Eurostat, 2024; National Center for Health Statistics, 2022; U.S. Census Bureau, 2020) show that regions of the same country at matched e_0 differ substantially in λ (Figure 1D). The within-country CV of λ among regions within a three-year e_0 band ranges from 0.10 to 0.36 (median 0.14), comparable to the cross-national Phase II CV (0.31). In Spain, regional λ in 2019 spans 5.1–14.3 births per 1,000 despite an e_0 spread of only 4.4 years. Longitudinal sub-populations from the Human Mortality Database (2024) likewise

diverge: East and West Germany follow different λ – e_0 paths, with a mean matched- e_0 gap of 2.5 births per 1,000 (27% of the mean).

Conclusion

None of this overturns the regularity Itao (2026) documents. It does relocate it. The first pathway is a timing-locked mortality–fertility co-movement that becomes sharper once each country is read on its own clock. The second pathway is better understood as heterogeneous convergence toward low fertility than as a single universal constant. Event-time alignment and hierarchical modelling are natural complements to the statistical-physics approach, pushing the same reasoning down to the within-country scales where policy operates.

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Conflicts of Interest

The author declares none.

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